Transferable Adversarial Attacks for Remote Sensing Object Recognition via Spatial-Frequency Co-Transformation

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***Abstract*— Adversarial attacks serve as an efficient approach to investigating model robustness, providing insights into internal weaknesses. In real-world applications, the model deployment typically adheres to a black-box setting, necessitating the transferability of adversarial examples crafted on a source model to others. Attack methods in the general computer vision field often employ global input transformations in individual spatial or frequency domains to boost adversarial transferability. However, the recognition of remote sensing objects primarily relies on target-related discriminative regions, whose determination exhibits significant model specificity. Besides, the coupling between objects and background further exacerbates the gap between models. Consequently, the transferability of adversarial examples is limited due to overfitting to the source model. To tackle this problem, we propose a spatial-frequency co-transformation (SFCoT) to improve adversarial transferability for remote sensing object recognition. Specifically, the input image is decomposed into blocks and components in the spatial and frequency domains, respectively. Then, a selective frequency transformation (SFT) is performed on the low-frequency components to narrow intermodel gaps. Subsequently, modular spatial transformations (MSTs) are adopted in blocks to enhance target-related diversity. Incorporating transformations across domains effectively mitigates the overfitting to model-specific information, leading to better adversarial transferability. Extensive experiments have been conducted on FGSCR-42 and MTARSI datasets, and the results demonstrate that the proposed method achieves state-of-the-art performance across various model architectures. The code will be released at https://github.com/fuyimin96/SFCoT.**

***Index Terms*— Adversarial attacks, adversarial transferability, model robustness, object recognition, remote sensing.**

I. Introduction

DEEP neural networks (DNNs) [1] have achieved remarkable success across multiple remote sensing tasks, such as scene understanding [2], data fusion [3], and image enhancement [4]. However, their deployment in real-world applications remains constrained. One primary reason is the vulnerability of DNNs to adversarial examples, which are crafted by adding imperceptible perturbations to original input images, misleading models to produce incorrect predictions. This vulnerability poses serious security concerns and underscores the necessity of improving model robustness. In response, extensive research on adversarial attacks [5] has been conducted to assess model robustness and identify internal weaknesses beforehand.

Existing adversarial attacks can be categorized into two types: white-box attacks [7], [8], [9], [10] and black-box attacks [11], [12], [13], [14], [15], [16], [17]. Under the white-box setting, the architecture and parameters of the target model are fully exposed to the attacker. On the contrary, the knowledge about the target model is inaccessible under the black-box setting, which is more in line with realistic model deployment. In this scenario, the adversarial examples crafted on a source model should be transferable to other target models. Within the field of general computer vision, various techniques have been adopted to enhance the transferability of adversarial examples, including momentum integration [11], input transformation [12], [13], [14], [15], importance estimation [16], and model ensemble [17]. Among these techniques, input transformation stands out due to its advantages in simplicity and flexibility. The overfitting to model-specific information is alleviated by simulating diverse input patterns for gradient calculation, thereby enhancing adversarial transferability.

Besides advancements in general computer vision, ongoing research on adversarial attacks is being conducted in the field of remote sensing. The concept of adversarial attacks was first introduced into the remote sensing community on the scene classification task [18]. Assessments of the threat posed by adversarial examples reveal the significance of model robustness and have attracted increasing research attention. Following this, further explorations of adversarial attacks on scene classification were carried out, with respect to data diversity [19], adversarial transferability [20], and multimodal collaboration [21]. Apart from scene classification, object detection [22] is another research focus of adversarial attacks in remote sensing. Current investigations on adversarial attacks against aerial detection are mainly focused on optical images, where attack efficacy is enhanced with consideration of physical practicability [23], landscape consistency [24], and visual stealthiness [25]. However, rare attention has been paid to remote sensing object recognition [26], [27], [28], on which

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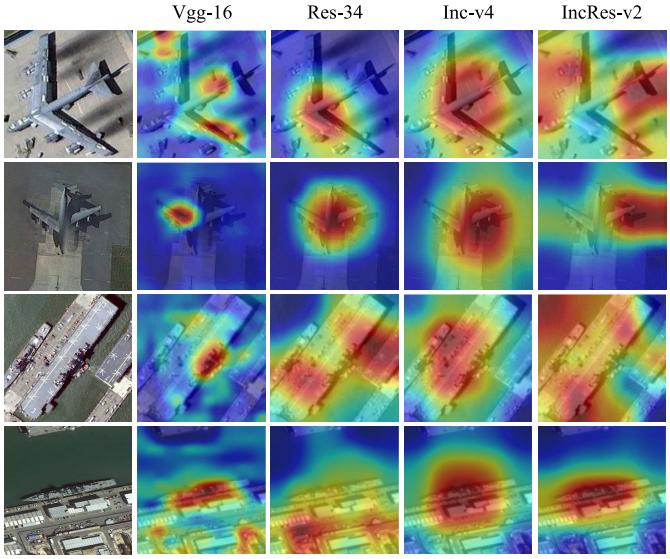


Fig. 1. Demonstration of differences in the focus on remote sensing object recognition across different models. We employ Grad-CAM [6] to obtain visual explanations of how different models pay attention to the object for recognition. All samples listed can be correctly recognized by each model, but with obvious differences in the focus of each model.

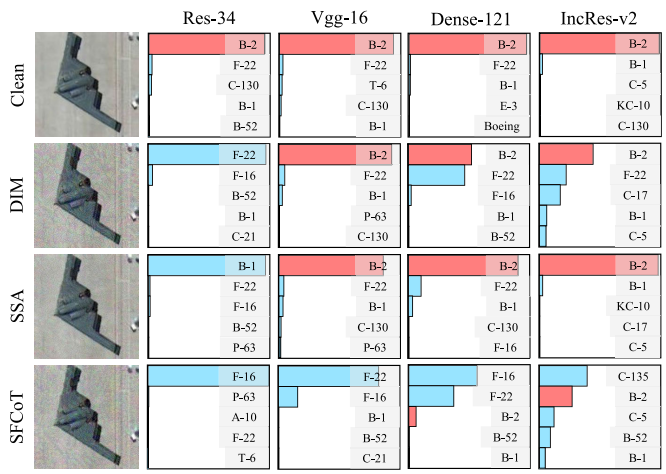


Fig. 2. Comparisons of different types of attacks. The clean image belongs to class **B-2**, which is colored in red among the top-5 confidence plots versus blue for the other classes. ResNet34 is employed as the source model to craft adversarial examples. The proposed method can successfully fool the models under both white-box and black-box settings.

there lacks a thorough validation across different experimental setups and model architectures.

As demonstrated in Fig. 1, the recognition of remote sensing objects primarily relies on target-related discriminative regions, whose determination exhibits significant differences across different models (i.e., model specificity). This complicates the coverage of potential target-related input patterns, resulting in the generation of adversarial examples suffering from overfitting to model-specific information. Additionally, objects in remote sensing images are strongly coupled with the background, which exacerbates the gap between different models due to background complexity. These factors collectively hinder the transferability of adversarial examples, leading to low black-box attack efficacy. Therefore, the key to achieving transferable adversarial attacks for remote sensing object recognition lies in the sufficient simulation of target-related input patterns and the effective decoupling of background information. However, existing attack methods typically apply global transformations in a single domain, falling short in meeting both key requirements.

To solve the problems above, we propose spatial-frequency co-transformation (SFCoT) to enhance the transferability of adversarial examples for remote sensing object recognition. Different from existing attack methods based on single-domain input transformations, SFCoT incorporates transformations in both the spatial and frequency domains. First, the input image is decomposed into image blocks and frequency components in a collaborative manner. Then, frequency transformation is selectively performed on the low-frequency components to decouple object and background information. Subsequently, spatial transformations are modularly applied to enhance target-related diversity in observation and environment changes. Finally, the set of transformed images are jointly used for gradient calculation. Fig. 2 illustrates a comparison of adversarial examples generated by SFCoT and single-domain transformations under both white-box and black-box settings. It can be seen that SFCoT leads to stronger adversarial transferability. We conduct extensive experiments on the FGSCR-42 [26] and MTARSI [27] datasets to validate the effectiveness of the proposed method, which achieves state-of-the-art performance across various model architectures under both white-box and black-box settings.

Our main contributions are summarized as follows.

1) We propose SFCoT, which incorporates transformations in both the spatial and frequency domains to improve the transferability of adversarial examples for remote sensing object recognition.
2) We propose a selective frequency transformation (SFT) on the low-frequency components. This effectively narrows intermodel gaps by decoupling the object from the background while retaining critical information.
3) We develop an adaptive partitioning strategy to divide the image into blocks, where spatial transformations are modularly applied to simulate diverse target-related input patterns in observation and environment changes.
4) We thoroughly benchmark adversarial attacks for remote sensing object recognition. Comprehensive evaluations of model robustness across various architectures provide deep insights into internal weaknesses when deploying DNNs in real-world applications.

The rest of this article is structured as follows. Section II reviews the works related to our work. Section III clarifies the problem setting and elaborates on the technical details of our proposed method. Section IV presents the experimental results and analysis. Finally, Section V concludes the work in this article and proposes directions for future research.

II. Related Works

In this section, we first review the related works in adversarial attacks and defenses, corresponding to the generation and resistance of adversarial examples, respectively. After that, the attack-defense race in remote sensing is presented.

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A. Adversarial Attacks

Existing adversarial attack methods can be categorized into two types: white-box attacks and black-box attacks.

Under the white-box setting, the adversarial examples are tailored toward the target model with access to its structure and parameters, yielding high success rates. The gradient information of loss function (with respect to the input) is commonly used to guide the generation process. Goodfellow et al. [7] proposed the fast gradient sign method (FGSM) to reduce computational consumption of box-constrained method [29] through single-step update. Kurakin et al. [9] extended FGSM to an iterative version, termed I-FGSM, which improves the white-box success rate at the cost of transferability. Similarly, Madry et al. [10] proposed project gradient descent (PGD), which is initialized from a random point and employs a projection operation to restrict the size of perturbation.

In contrast, knowledge of the target model is absent under the black-box setting. Some early black-box attack methods approximate the information of target models by queries [30], [31], which is impractical due to huge computational costs. Therefore, recent studies focus on launching black-box attacks by improving the transferability of adversarial examples. Dong et al. [11] integrated the momentum into the iterative generation process to escape from local maxima. For more stable update, Wang et al. [32] enhanced the momentum by including the average gradient of the previous iterations into account. Liu et al. [17] proposed ensemble-based attack to enhance the cross-model transferability. Instead of simple prediction integration, Xiong et al. [33] reduced the gradient variance of ensemble models to enhance the generalizability of crafted adversaries. With the aim of distorting important features, Wang et al. [16] proposed feature importance-aware attack (FIA) to mitigate overfitting to model-specific features. Zhang et al. [34] adopted neuron attribution for more accurate importance measurement. In addition to the aforementioned techniques, input transformations are widely adopted for transferability enhancement, aligning closely with our main focus in this article. Xie et al. [12] first investigated the promotion of transferability with respect to input diversity, which involves random resizing and padding of the image. Dong et al. [13] proposed a translation-invariant attack method (TIM) to reduce the sensitivity to discriminative regions of the source model. Lin et al. [14] discovered the scale-invariant property of DNNs and proposed a scale-invariant attack method (SIM) to generate more transferable adversarial examples. Gao et al. [35] proposed a patch-wise iterative attack method (PIM) to improve the transferability. Beyond transformations on a single image, Wang et al. [36] proposed Admix, which includes information from other classes into the gradient calculation. Long et al. [15] proposed a spectrum simulation attack (SSA) to narrow intermodel gaps in the frequency domain. Recently, Wang et al. [37] proposed block shuffle and rotation (BSR) to improve transferability by disrupting model attentions.

B. Adversarial Defenses

To mitigate the threat posed by adversarial examples, various adversarial defense methods have been proposed to improve model robustness, primarily through two approaches: adversarial training and preprocessing.

Adversarial training [38] includes adversarial examples into the training process, thereby making the model resistant to attacks. Madry et al. [10] investigated adversarial training from a robust optimization perspective. Tramèr et al. [39] proposed ensemble adversarial training, which decouples the generation of adversarial examples from the model to boost defense against attacks. Xie et al. [40] Wong et al. [41] combined single-step FGSM with random initialization to perform adversarial training in a low-cost computational manner. However, the applicability of adversarial training to large-scale datasets and complex network architectures is still limited.

Shifting the focus away from the model training process, various forms of preprocessing are employed to eliminate adversarial perturbations from the input data before feeding into models. Guo et al. [42] applied multiple image transformations to increase model-agnostic defense efficacy. Liao et al. [43] employed the difference in high-level representation to guide the denoising of adversarial examples. Liu et al. [44] designed a JPEG-based defensive compression framework, which rectifies adversarial examples by feature distillation. Jia et al. [45] adopted randomized smoothing to generalize certified robustness to top-k predictions. Faced with continuously evolving attacks, Naseer et al. [46] adversarially trained a neural representation purifier to learn optimal input processing in a self-supervised manner. In addition to single-image-based scenarios, some recent studies have enhanced model robustness against interference through multisource information fusion [47], [48].

C. Attack–Defense Race in Remote Sensing

Besides the general computer vision field, adversarial attacks and defenses have also drawn significant attention in remote sensing, presenting an ongoing attack-defense race.

Xu et al. [18] systematically assessed the threat of adversarial examples on DNNs for remote sensing scene classification. Subsequently, Chen et al. [19] investigated the transferability of different attacks, providing a more thorough empirical study in scene classification. Xu and Ghamisi [20] first explored universal adversarial examples in remote sensing by implementing Mixup-attack at the feature level, which is simple but effective. Shi et al. [21] attacked multimodal remote sensing classification networks through modality-specific generators. Taking into account the intrinsic characteristics of remote sensing images, Su et al. [49] proposed a defense framework based on reconstruction-assisted and distance-optimized adversarial training. Li et al. [50] integrated low-rank representation with deep learning techniques to improve model robustness against anomalies in hyperspectral images. In addition to scene-level interpretation, research toward target-level understanding has also been conducted. Lian et al. [23] established the attack benchmark of adversarial patches for aerial object detection. Sun et al. [24] leveraged the first-order difference to facilitate the patch selection for attack. Tang et al. [25] generated natural weather-style perturbations to fool optical aerial detectors. Chen and Chu [51] proposed an adaptive

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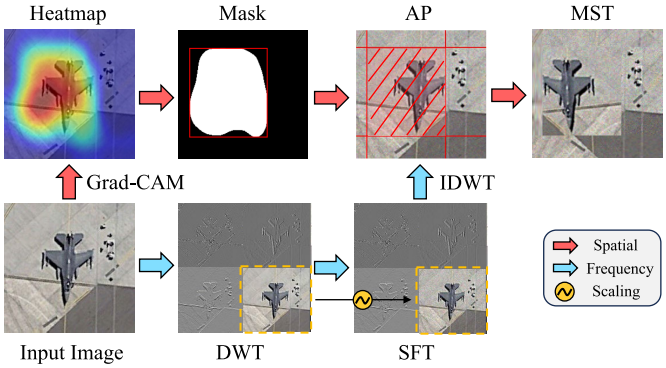


Fig. 3. Transformation pipeline of SFCoT for gradient calculation.

defense pipeline to improve model robustness of aerial detection in the adversarial environment. Some recent attempts have been made on the task of remote sensing object recognition. Zhang et al. [52] generated synthetic aperture radar (SAR) adversarial images according to the characteristics of SAR data. Peng et al. [53] leveraged the scattering model to guide adversarial attack for SAR automatic target recognition, and employed their proposal to defend against malicious scatterers. Nevertheless, existing research on adversarial attacks and defenses for object recognition primarily focuses on SAR images under the white-box setting with limited classes.

III. Methodology

Given an input image, transformations in both spatial and frequency domains are collaboratively employed to simulate diverse input patterns. In the frequency domain, the image is decomposed into different subbands, where amplitude scaling is selectively applied to the low-frequency components. In the spatial domain, the image reconstructed from the frequency domain is adaptively divided into blocks. Spatial transformations are then modularly performed based on the content of each block. The pipeline of SFCoT is illustrated in Fig. 3. In this section, we first introduce the problem formulation of input transformation-based adversarial attacks under the black-box setting. Then, we elaborate on the technical details of improving adversarial transferability with SFCoT.

A. Problem Formulation

Formally, given a target model $f_\theta : x \to y$ for object recognition, where x, y, and θ refer to the input image, true class label, and model parameters. The aim of adversarial attacks is to generate an adversarial example $\bar{x}$ by adding an imperceptible perturbation on x, thereby fooling the model to make false prediction, i.e., $f_\theta(\bar{x}) \neq y$. To control the magnitude of the perturbation, the adversarial example is normally confined inside a $\ell_p -$ norm ball centered at x with radius ϵ. Following previous works [11], [12], [13], [14], [15], [16], we use $\ell_\infty -$ norm in this article, and the generation of adversarial examples can be formulated as a constrained optimization problem

$$\arg\max_{\bar{x}} J\left(\bar{x}, y\right), \quad \text{s.t.} \quad \left\|x - \bar{x}\right\|_\infty \leq \epsilon \tag{1}$$

where $J(\cdot,\cdot)$ is the loss function of the classifier (i.e., cross-entropy loss), and the single-step gradient-based update process can be expressed as

$$\bar{x} = x + \epsilon \cdot \text{sign}(\nabla_x J(x, y, \theta)) \tag{2}$$

where ∇_x refers to the gradient with respect to the input. To achieve higher attack efficacy, the generation process is typically carried out in an iterative form

$$\bar{x}_{t+1} = \text{Clip}_{x,\epsilon}\left\{\bar{x}_t + \alpha \cdot \text{sign}\left(\nabla_{\bar{x}_t} J\left(\bar{x}_t, y, \theta\right)\right)\right\} \tag{3}$$

where t and α represent the iteration number and step size, respectively. $\text{Clip}_{x,\epsilon}\{\cdot\}$ refers to the pixel-wise clipping operation. However, parameters θ of the target model is inaccessible under the black-box setting. In this case, a source model f_ϕ with accessible model parameters ϕ is often used as the surrogate, which generates transferable adversarial examples for target models. As an effective way to improve such transferability, various transformations $\mathcal{T}(\cdot)$ are applied to the input at each iteration for gradient calculation:

$$\bar{x}_{t+1} = \text{Clip}_{x,\epsilon}\left\{\bar{x}_t + \alpha \cdot \text{sign}\left(\nabla_{\bar{x}_t} J\left(\mathcal{T}\left(\bar{x}_t\right), y, \phi\right)\right)\right\}. \tag{4}$$

B. Collaborative Image Decomposition

Whether in the spatial or frequency domain, existing methods typically apply a single transformation to the entire input image. However, this can limit the improvement of transferability to adversarial attacks for remote sensing object recognition. On the one hand, potential target-related input patterns cannot be comprehensively covered through a single spatial transformation. On the other hand, excessive focus on redundant information from the background in high-frequency components is inefficient for narrowing intermodel gaps.

To get rid of this limitation, we decompose the input image in the spatial and frequency domains in a collaborative manner. First, we apply 2-D discrete wavelet transform (DWT) to decompose the input image x into components in four frequency subbands:

$$\{x_{LL}, x_{LH}, x_{HL}, x_{HH}\} = \text{DWT}(x) \tag{5}$$

where L and H denote the low-pass and high-pass filters, respectively. Compared to other high-frequency components, the low-frequency components x_{LL} primarily focuses on retaining critical information about the object rather than texture and edge details. Consequently, the object and background information are implicitly decoupled in the low-frequency subband, making the transformation therein more conducive to narrowing intermodel gaps. Meanwhile, in the spatial domain, we develop an adaptive partition (AP) strategy to divide the image into blocks. Specifically, the heatmap $\mathcal{M}$ is calculated using gradient-weighted class activation mapping (Grad-CAM) to reflect the attention of source model to the input image

$$\mathcal{M} = \text{ReLU}\left(\sum_{k \in K} \alpha_k^y A^k\right) \tag{6}$$

where K is the number of channels in the last convolutional layer. For each channel k, the corresponding feature map and

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neuron importance weights to the true class are denoted as A^k and α_k^y, respectively. Then, we upsample $\mathcal{M}$ to the resolution of the input image and convert it to a binary mask $\tilde{\mathcal{M}}$ by thresholding the value at each location

$$\tilde{\mathcal{M}}_{i,j} = \begin{cases} 1, & \text{if } \mathcal{M}_{i,j} \geq \tau \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where i and j are coordinate indices in the width and height dimensions, and τ is the activation threshold. Subsequently, we delineate a bounding box B to cover positive values in $\mathcal{M}$

$$B = [\min i, \max i] \times [\min j, \max j] \tag{8}$$

where the minimum and maximum indices are taken over all (i, j) such that $\tilde{\mathcal{M}}_{i,j} > 0$, and $\times$ denotes the Cartesian product. Finally, the inverse DWT (IDWT) is applied to the frequent components, from which the reconstructed image $\tilde{x}$ is divided into $P+1$ blocks according to the coordinates of B

$$\mathrm{AP}(\tilde{x}) = \{\tilde{x}_o, \tilde{x}_e^1, \ldots, \tilde{x}_e^P\} \tag{9}$$

where the inner block $\tilde{x}_o$ is considered to be target-related, while the blocks, $\tilde{\mathbf{x}}_e = \{\tilde{x}_e^p\}_{p=1}^P$, relate to the environment. To further enhance target-related diversity, the dimensions of $\tilde{x}_o$ are randomly adjusted within a range of η at each iteration.

C. Selective Frequency Transformation

The high-frequency components of the image typically contain edge and texture details, to which the model inherently exhibits stronger robustness [54]. As a result, performing transformations on high-frequency components is inefficient for narrowing intermodel gaps. In contrast, low-frequency components primarily focus on capturing critical information. Since the decoupling between object and background is implicitly realized, performing transformations on low-frequency components can better enhance adversarial transferability.

To this end, we selectively perform the transformation on low-frequency components without altering the high-frequency components. Specifically, we adopt a random amplitude scaling operation on x_{LL} to vary the values of each pixel

$$\tilde{x}_{LL} = x_{LL} \odot \mathcal{R} \tag{10}$$

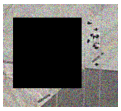
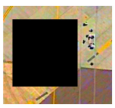
where $\odot$ is the Hadamard product, and $\mathcal{R} \sim U(1-\delta, 1+\delta)$ is the matrix of scaling factors for each pixel. The overall scaling amplitude is controlled by hyperparameter δ. Then, the scaled low-frequency components are combine with the unaltered high-frequency components to obtain the reconstructed image $\tilde{x}$ using IDWT. The overall process of SFT can be expressed as

$$\mathcal{T}_f(x) = \mathrm{IDWT}(\tilde{x}_{LL}, x_{LH}, x_{HL}, x_{HH}). \tag{11}$$

To efficiently incorporate transformations across different domains, spatial transformations are subsequently applied to the reconstructed image instead of the original input image. The benefit of following such a pipeline is that narrowing the intermodel gap can facilitate the comprehensive coverage of potential input patterns.

TABLE I

ILLUSTRATION OF SOME TRANSFORMATIONS APPLIED TO DIFFERENT TYPES OF IMAGE BLOCKS IN THE SPATIAL DOMAIN

Resize	Flip	Scale	Rotation	Shift
				
Noise	**Dropout**	**Brightness**	**Contrast**	**Saturation**
				

D. Modular Spatial Transformation

The determination of target-related discriminative regions exhibits significant model specificity. When generating adversarial examples, overfitting to the source model is exacerbated, thereby hindering the transferability across models. As discussed above, the simulation of target-related input patterns through a single global spatial transformation is insufficient. On the premise of preserving semantic information, target-related diversity can be enhanced by combining various transformations. Consequently, overfitting to model-specific information can be effectively mitigated.

Motivated by this insight, we modularly apply various transformations to different image blocks in the spatial domain. Specifically, the selection scope of transformations applied to each block is specified depending on the corresponding content (i.e., target or environment). For the target-related block $\tilde{x}_o$, the involved transformations $\mathcal{T}_o(\cdot)$ primarily focus on simulating changes in observation, part of which are illustrated in the top row of Table I. Additionally, changes in the environment are simulated by applying transformations $\mathcal{T}_e(\cdot)$ to the remaining blocks, as shown in the bottom row of Table I. In each iteration of generating adversarial examples, one transform is randomly selected from the corresponding scope for each image block and applied accordingly. The combination of spatial transformations across different blocks significantly enhances target-related diversity, and the process of modular spatial transformation (MST) can be expressed as

$$\mathcal{T}_s(\tilde{x}) = \mathcal{T}_o(\tilde{x}_o) \oplus \{\mathcal{T}_e^n(\tilde{x}_e^p)\}_{p=1}^P \tag{12}$$

where $\oplus$ refers to the operation of stitching different transformed blocks into a whole image.

E. Co-Transformation Attack Algorithm

Formally, the integrated transformation process of SFCoT can be expressed as

$$\mathcal{T}_{Co}(x) = \mathcal{T}_s(\mathrm{AP}(\mathcal{T}_f(x))). \tag{13}$$

Similar to the existing attack methods based on input transformation, SFCoT can be integrated into each iteration of the gradient-based generation process of adversarial examples

$$\bar{x}_{t+1} = \mathrm{Clip}_{x,\epsilon}\{\bar{x}_t + \alpha \cdot \mathrm{sign}(\nabla_{\bar{x}_t} J(\mathcal{T}_{Co}(\bar{x}_t), y, \phi))\}. \tag{14}$$

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Algorithm 1 SFCoT Attack Algorithm

Input: A source model f_ϕ with loss function J, and a input image x with class label y; The size of transformed image set N, range of dimension η, and the range of scaling δ; The constraint on perturbation ϵ, step size α, number of iterations T, and decay factor μ.

Output: The adversarial example $\bar{x}$.

Initialize: $g_0 = 0$, $\bar{x}_0 = x$.
for $t = 0$ **to** $T - 1$ **do**
  for $n = 1$ **to** N **do**
    Get the transformed image by Eq. (13);
  end for
  Calculate the average gradient $\bar{g}_{t+1}$ of the set of transformed images by Eq. (15);
  Update the momentum by Eq. (16);
  Update the generated adversarial example $\bar{x}_{t+1}$ based on the gradient sign by Eq. (17);
end for
return $\bar{x} = \bar{x}_T$

For a more stable update, the gradient is calculated on a set of transformed images rather than individual transformed images, along with the integration of the momentum term

$$\bar{g}_{t+1} = \frac{1}{N}\sum_{n=1}^{N} \nabla_{\bar{x}_t^n} J\left(\mathcal{T}_{Co}\left(\bar{x}_t^n\right), y, \phi\right) \tag{15}$$

$$g_{t+1} = \mu g_t + \frac{\bar{g}_{t+1}}{\left\|\bar{g}_{t+1}\right\|_1} \tag{16}$$

where N is the size of the set of transformed images, and μ is the decay factor. In this case, the generation process elaborated in (14) is transformed into

$$\bar{x}_{t+1} = \mathrm{Clip}_{x,\epsilon}\left\{\bar{x}_t + \alpha \cdot \mathrm{sign}(g_{t+1})\right\}. \tag{17}$$

The overall procedure of the attack algorithm that employs the proposed SFCoT to iteratively generate adversarial examples is summarized in Algorithm 1.

IV. Experiments

A. Experimental Setup

1) Dataset: We conduct experiments on the FGSCR-42 and MTARSI datasets, which contain 42 and 20 types of ships and aircraft in remote sensing images, respectively. Some samples from each dataset are shown in Fig. 4, and all images are resized to 224 × 224.

2) Model Architectures: We choose six representative DNNs to evaluate the attack performance, including VGG16 (Vgg-16) [55], VGG19 (Vgg-19) [55], ResNet-34 (Res-34) [56], ResNet-50 (Res-50) [56], DenseNet-121 (Dense-121) [57], and Inception-ResNet-V2 (IncRes-v2) [58]. Each network is employed in turn as the source model to generate adversarial examples against all the six models.



(a)

(b)

Fig. 4. Some image samples taken from the FGSCR-42 and MTARSI datasets. (a) FGSCR-42. (b) MTARSI.

3) Comparison Methods: To prove the effectiveness of the proposed method, we compare SFCoT with various state-of-the-art attack methods, including PGD [10], MIM [11], TIM [13], PIM [35], DIM [12], SIM [14], Admix [36], Mixup/Mixcut [20], SSA [15], and BSR [37].

4) General Implementation: In our experiments, the target models are normally trained using the entire training data. Only test data that can be correctly classified by all models are chosen to evaluate attack performance. During the generation of adversarial examples, the constraint on perturbation ϵ, the step size α, and the number of iterations T are set to 16, 1, and 30, respectively. The parameters τ, η, and δ involved in the transformation process of SFCoT are set to 0.5, 0.4, and 0.1, respectively. Parameter settings specific to each comparison method adhere to their recommended configurations. The size of the transformed image set N for gradient calculation is set to 5. The experiments are conducted in the PyTorch framework with Nvidia RTX 4090 GPU support.

B. Performance Comparison

1) Quantitative Results: We thoroughly evaluate the attack performance across different models under both white-box and black-box settings. The performance is evaluated by the attack success rate, i.e., the proportion of generated adversarial examples that successfully mislead the target model into incorrect classifications. The results are shown in Tables II and III, with source models and target models listed in rows and columns, respectively.

According to the results, the attack performance of SFCoT consistently surpasses that of other attack methods across models on each dataset. First, the proposed attack method achieves 100% attack success rate on the source model itself, indicating the enhancement of adversarial transferability without compromising white-box attack performance. Then, under the black-box setting, SFCoT significantly outperforms other methods in terms of adversarial transferability. For instance, most methods can achieve perfect white-box attack success

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TABLE II

ATTACK SUCCESS RATES ON THE FGSCR-42 DATASET AGAINST DIFFERENT MODELS. THE BEST ATTACK PERFORMANCE ON EACH MODEL IS HIGHLIGHTED IN BOLD

Model	Attack	Vgg-16	Vgg-19	Res-34	Res-50	Dense-121	IncRes-v2
Vgg-16	PGD	99.9%	94.7%	61.3%	65.2%	59.9%	48.3%
	MIM	99.7%	99.1%	78.1%	74.1%	77.7%	55.8%
	PIM	99.9%	99.5%	81.6%	78.9%	82.5%	56.2%
	TIM	99.9%	95.6%	75.9%	69.0%	72.3%	43.2%
	DIM	99.7%	99.5%	84.2%	80.3%	85.2%	59.1%
	SIM	99.9%	98.7%	79.7%	78.9%	81.0%	54.6%
	Admix	99.9%	99.7%	85.2%	83.8%	86.1%	56.5%
	Mixup	97.5%	93.7%	80.2%	78.4%	80.2%	56.9%
	Mixcut	99.9%	99.7%	82.2%	82.1%	85.4%	55.3%
	SSA	99.7%	99.1%	84.3%	82.9%	85.1%	59.7%
	BSR	**100.0%**	**100.0%**	91.1%	89.7%	91.7%	57.5%
	SFCoT	**100.0%**	**100.0%**	**96.8%**	**95.9%**	**97.0%**	**62.7%**
Vgg-19	PGD	91.0%	99.9%	53.5%	59.7%	52.9%	47.9%
	MIM	96.6%	99.4%	73.1%	73.5%	74.5%	56.3%
	PIM	98.4%	99.9%	78.4%	78.0%	79.4%	55.3%
	TIM	94.4%	99.9%	73.2%	66.8%	69.0%	43.0%
	DIM	98.9%	99.7%	80.1%	76.9%	81.3%	58.3%
	SIM	96.1%	99.8%	77.4%	75.8%	74.8%	54.2%
	Admix	98.3%	99.9%	81.2%	79.3%	79.3%	55.4%
	Mixup	98.3%	99.2%	80.8%	77.8%	78.2%	54.2%
	Mixcut	98.1%	99.1%	80.2%	77.8%	78.4%	54.5%
	SSA	97.8%	99.6%	76.8%	76.0%	76.3%	57.7%
	BSR	99.6%	99.9%	87.3%	85.9%	88.7%	57.0%
	SFCoT	**99.8%**	**100.0%**	**91.8%**	**91.7%**	**93.9%**	**58.5%**
Res-34	PGD	25.9%	19.4%	**100.0%**	52.5%	42.8%	51.0%
	MIM	70.5%	48.3%	99.9%	78.1%	72.3%	62.6%
	PIM	64.4%	43.4%	99.9%	78.9%	71.8%	59.1%
	TIM	24.5%	15.5%	99.9%	50.9%	47.9%	27.9%
	DIM	81.6%	64.9%	99.9%	86.7%	84.0%	65.2%
	SIM	78.0%	64.3%	99.9%	83.8%	86.0%	59.7%
	Admix	82.2%	70.8%	99.9%	88.3%	89.0%	62.4%
	Mixup	77.8%	64.2%	97.5%	84.9%	86.1%	60.3%
	Mixcut	77.9%	63.7%	97.4%	85.0%	86.3%	60.4%
	SSA	82.7%	68.9%	**100.0%**	88.9%	85.5%	63.2%
	BSR	89.6%	80.3%	**100.0%**	96.3%	93.3%	66.4%
	SFCoT	**93.4%**	**86.9%**	**100.0%**	**98.7%**	**97.6%**	**68.3%**
Res-50	PGD	24.8%	18.7%	47.2%	99.9%	46.5%	50.4%
	MIM	64.0%	42.0%	79.5%	99.9%	74.6%	61.4%
	PIM	59.8%	40.2%	82.7%	99.9%	75.6%	57.9%
	TIM	25.3%	14.6%	60.7%	99.9%	55.6%	27.5%
	DIM	78.8%	63.0%	90.6%	99.9%	91.6%	64.2%
	SIM	76.9%	62.4%	87.7%	99.9%	88.6%	58.5%
	Admix	81.3%	69.8%	94.5%	99.9%	94.9%	61.3%
	Mixup	76.4%	61.0%	89.5%	97.7%	91.2%	59.2%
	Mixcut	76.5%	60.7%	89.1%	99.9%	91.4%	59.1%
	SSA	81.0%	68.1%	92.3%	99.9%	89.7%	62.2%
	BSR	84.1%	72.5%	97.1%	99.9%	98.4%	65.9%
	SFCoT	**90.6%**	**85.0%**	**99.0%**	**100.0%**	**99.6%**	**66.0%**
Dense-121	PGD	26.5%	20.2%	42.7%	52.0%	**100.0%**	50.5%
	MIM	69.4%	48.8%	77.9%	75.7%	99.9%	59.3%
	PIM	67.6%	50.2%	80.6%	80.8%	**100.0%**	56.7%
	TIM	28.1%	17.8%	57.6%	55.4%	99.9%	27.9%
	DIM	79.6%	67.0%	86.3%	87.6%	99.9%	62.1%
	SIM	74.3%	62.1%	81.4%	78.6%	**100.0%**	57.3%
	Admix	79.8%	68.8%	87.8%	86.4%	99.9%	60.2%
	Mixup	74.9%	62.4%	83.8%	81.8%	99.9%	59.6%
	Mixcut	74.7%	62.5%	84.1%	81.4%	99.9%	59.3%
	SSA	82.3%	73.4%	87.6%	87.7%	**100.0%**	60.1%
	BSR	84.2%	74.4%	92.6%	94.3%	99.9%	**65.3%**
	SFCoT	**87.3%**	**81.2%**	**97.1%**	**98.3%**	**100.0%**	65.0%
IncRes-v2	PGD	20.1%	15.3%	27.6%	36.8%	31.6%	**100.0%**
	MIM	23.3%	16.3%	33.2%	38.5%	37.3%	**100.0%**
	PIM	25.4%	16.7%	36.3%	41.3%	39.8%	**100.0%**
	TIM	9.7%	5.2%	27.0%	24.6%	23.6%	99.9%
	DIM	39.8%	22.6%	52.6%	53.3%	55.1%	**100.0%**
	SIM	27.9%	15.7%	34.8%	40.4%	41.5%	96.0%
	Admix	28.9%	16.1%	38.5%	43.4%	44.9%	99.2%
	Mixup	20.4%	14.1%	29.1%	34.6%	33.9%	97.3%
	Mixcut	20.3%	14.3%	29.0%	33.8%	33.7%	97.1%
	SSA	28.7%	17.1%	39.5%	43.0%	41.6%	99.9%
	BSR	45.9%	**24.4%**	56.6%	59.6%	**60.7%**	**100.0%**
	SFCoT	**46.8%**	23.5%	**58.3%**	**59.7%**	58.7%	**100.0%**

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TABLE III

ATTACK SUCCESS RATES ON THE MTARSI DATASET AGAINST DIFFERENT MODELS. THE BEST ATTACK PERFORMANCE ON EACH MODEL IS HIGHLIGHTED IN BOLD

Model	Attack	Vgg-16	Vgg-19	Res-34	Res-50	Dense-121	IncRes-v2
Vgg-16	PGD	**100.0%**	64.7%	20.3%	22.5%	28.3%	8.4%
	MIM	99.9%	93.6%	47.8%	51.3%	60.4%	29.7%
	PIM	**100.0%**	93.4%	48.5%	53.0%	58.8%	25.2%
	TIM	**100.0%**	87.2%	45.4%	50.7%	52.8%	19.5%
	DIM	**100.0%**	97.3%	61.1%	65.8%	69.2%	32.8%
	SIM	**100.0%**	96.6%	47.5%	53.5%	63.2%	30.4%
	Admix	**100.0%**	98.6%	56.8%	60.4%	68.8%	32.1%
	Mixup	99.8%	97.9%	50.8%	54.6%	66.0%	30.6%
	Mixcut	97.3%	98.2%	53.1%	57.4%	66.4%	30.7%
	SSA	**100.0%**	95.3%	51.5%	55.2%	62.7%	28.3%
	BSR	**100.0%**	99.4%	65.6%	70.1%	78.0%	33.2%
	SFCoT	**100.0%**	**99.9%**	**77.1%**	**81.9%**	**87.2%**	**34.7%**
Vgg-19	PGD	61.3%	**100.0%**	18.9%	20.3%	27.9%	7.8%
	MIM	94.8%	99.9%	48.3%	52.5%	59.8%	25.4%
	PIM	94.7%	**100.0%**	50.7%	51.5%	58.0%	22.2%
	TIM	85.2%	**100.0%**	44.8%	52.1%	52.6%	15.5%
	DIM	98.1%	99.9%	59.0%	63.6%	69.6%	30.4%
	SIM	97.8%	**100.0%**	52.0%	58.2%	63.9%	25.1%
	Admix	99.7%	**100.0%**	62.0%	64.9%	70.9%	26.2%
	Mixup	98.6%	99.1%	55.5%	60.3%	67.0%	24.5%
	Mixcut	98.9%	99.4%	54.7%	59.2%	67.3%	25.1%
	SSA	96.7%	99.9%	52.6%	53.8%	63.4%	27.7%
	BSR	99.8%	**100.0%**	66.6%	71.1%	77.6%	30.5%
	SFCoT	**100.0%**	**100.0%**	**76.7%**	**81.5%**	**88.5%**	**31.2%**
Res-34	PGD	4.1%	3.8%	75.8%	12.3%	9.3%	7.4%
	MIM	27.0%	27.9%	**100.0%**	45.4%	47.9%	31.2%
	PIM	18.6%	20.2%	99.9%	40.9%	41.1%	20.1%
	TIM	3.1%	2.4%	**100.0%**	19.6%	16.3%	2.7%
	DIM	44.9%	45.5%	99.9%	68.4%	68.3%	33.8%
	SIM	38.5%	38.9%	99.8%	48.1%	51.3%	25.8%
	Admix	49.4%	49.1%	**100.0%**	64.3%	66.2%	31.3%
	Mixup	40.9%	40.8%	92.9%	55.4%	60.4%	28.3%
	Mixcut	38.2%	40.9%	99.7%	55.4%	59.4%	28.5%
	SSA	51.4%	51.8%	99.9%	65.5%	64.6%	29.8%
	BSR	68.7%	68.0%	**100.0%**	88.8%	84.6%	**35.8%**
	SFCoT	**77.2%**	**75.5%**	**100.0%**	**91.6%**	**86.0%**	32.4%
Res-50	PGD	4.9%	5.0%	11.5%	66.6%	8.7%	7.3%
	MIM	22.1%	23.4%	41.2%	99.8%	45.9%	25.9%
	PIM	15.8%	17.1%	37.7%	99.2%	39.0%	17.1%
	TIM	2.7%	2.9%	16.0%	99.0%	15.9%	1.7%
	DIM	42.9%	43.0%	67.4%	99.8%	71.6%	28.0%
	SIM	31.2%	29.2%	38.7%	93.4%	46.2%	21.3%
	Admix	43.0%	42.2%	55.7%	99.6%	66.7%	25.9%
	Mixup	34.6%	34.9%	46.3%	94.7%	57.8%	23.7%
	Mixcut	33.7%	34.3%	45.1%	98.6%	57.4%	23.6%
	SSA	50.1%	52.2%	65.6%	99.7%	64.6%	23.7%
	BSR	59.8%	61.0%	81.2%	**100.0%**	82.2%	28.3%
	SFCoT	**76.0%**	**75.7%**	**91.1%**	**100.0%**	**89.6%**	**28.6%**
Dense-121	PGD	5.4%	5.2%	12.6%	12.7%	78.3%	8.2%
	MIM	26.9%	28.1%	38.0%	38.6%	99.9%	28.9%
	PIM	21.8%	22.0%	37.5%	40.0%	99.9%	19.2%
	TIM	3.4%	3.5%	14.5%	18.9%	99.9%	3.0%
	DIM	48.0%	49.1%	61.6%	63.4%	99.9%	33.4%
	SIM	32.0%	33.7%	37.8%	41.6%	99.8%	25.2%
	Admix	43.9%	47.9%	54.9%	57.7%	**100.0%**	30.8%
	Mixup	34.8%	38.5%	45.1%	49.4%	96.0%	28.3%
	Mixcut	34.4%	37.8%	46.3%	47.5%	91.6%	28.0%
	SSA	49.1%	55.1%	60.3%	63.9%	99.9%	29.2%
	BSR	64.3%	64.2%	73.2%	79.7%	**100.0%**	**35.0%**
	SFCoT	**81.7%**	**82.3%**	**88.4%**	**91.9%**	**100.0%**	33.0%
IncRes-v2	PGD	2.8%	2.7%	5.3%	6.2%	4.6%	58.7%
	MIM	4.3%	5.0%	7.8%	8.8%	7.6%	**100.0%**
	PIM	4.7%	4.9%	8.0%	9.1%	8.5%	**100.0%**
	TIM	1.3%	1.1%	3.3%	4.2%	3.6%	**100.0%**
	DIM	6.6%	7.8%	12.8%	15.6%	13.9%	**100.0%**
	SIM	4.3%	5.4%	6.2%	8.3%	7.6%	62.3%
	Admix	4.9%	5.5%	8.5%	9.7%	8.4%	87.4%
	Mixup	3.7%	4.3%	6.9%	6.7%	7.2%	96.5%
	Mixcut	4.1%	4.3%	6.8%	6.5%	7.4%	96.6%
	SSA	4.9%	5.5%	9.0%	10.2%	8.6%	98.1%
	BSR	7.9%	9.5%	13.5%	16.7%	14.2%	**100.0%**
	SFCoT	**9.7%**	**11.2%**	**13.6%**	**19.1%**	**17.8%**	**100.0%**

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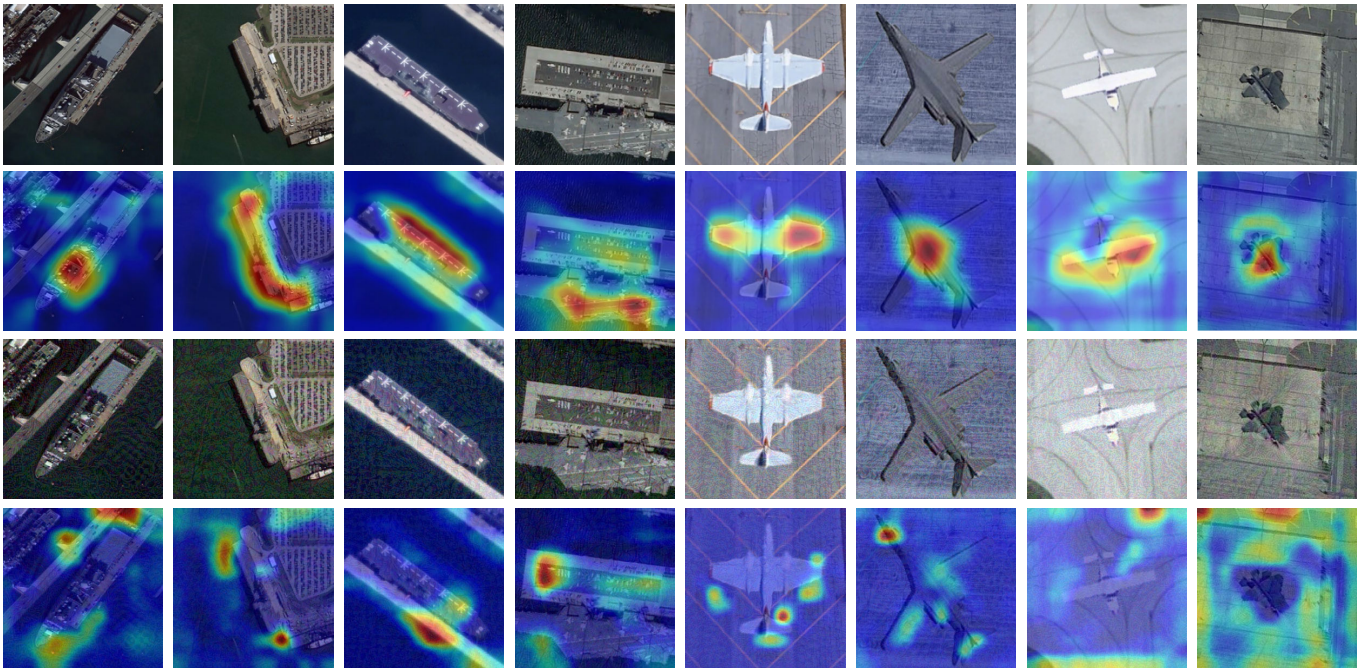


Fig. 5. Visualization for attention shift of the target model. The clean images and their attention maps are shown in the top two rows, with the corresponding adversarial examples generated by SFCoT and their attention maps shown in the bottom two rows.

rates against VGG models on the MTARSI dataset. However, when testing on different model architectures, SFCoT outperforms other methods by a clear margin of over 12% on average. In addition, we also observe that the transferability based on IncRes-v2 is markedly inferior to that of other model architectures (VGG and ResNet), irrespective of being used as the source or target model. This disparity is principally caused by the multiscale feature extraction of the inception module, implying the impact of architectural heterogeneity on model robustness. Furthermore, the overall attack success rate (OASR) on the MTARSI dataset is lower compared to the FGSCR-42 dataset. Such superiority in resistance to attacks is mainly attributed to the simulation of target and background combinations when constructing the MTARSI dataset.

2) Qualitative Results: To better illustrate how transferable attacks for remote sensing object recognition are achieved, we conduct a qualitative analysis of the proposed method. Specifically, Res-34 is used as the source model to generate adversarial examples on the MTARSI and FGSCR-42 datasets, while Vgg-16 serves as the target model to evaluate the attack efficacy. Visualization results of original input images and the generated adversarial examples, along with the corresponding attention maps, are shown pairwise in Fig. 5. According to the results, the visual discrepancy between the original input images and the adversarial examples is imperceptible. However, the attention of the target model is effectively shifted from target-related discriminative regions to irrelevant regions in the image. Consequently, the model is misled to make incorrect predictions.

3) Computational Complexity: Besides quantitative and qualitative analysis of the attack performance, we also compare the computational complexity of SFCoT with other input transformation-based attack methods. The experiments are conducted on the MTARSI dataset by using Res-34 as the source model. The results are reported by the time consumption of generating adversarial examples and the OASR across different target models. According to the results shown in Table IV, SFCoT can achieve leading attack performance with low computational complexity.

TABLE IV
ANALYSIS OF COMPUTATIONAL COMPLEXITY

Methods	SIM	Admix	SSA	BSR	SFCoT
Time (s)	785	2075	453	214	399
OASR	40.5	52.0	52.6	69.1	72.5

C. Ablation Studies

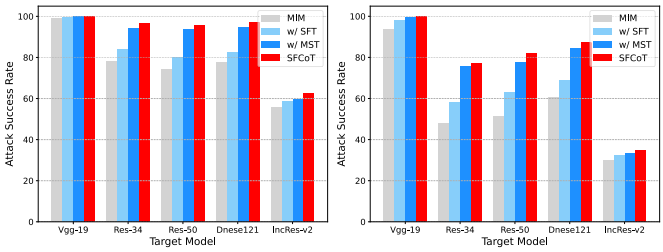
1) Effectiveness of Transformations in Each Domain: The overall input transformation process of the proposed method consists of two main subprocesses: the selective frequency transformation and the MST. We conduct ablation experiments on the FGSCR-42 and MTARSI datasets to demonstrate the effectiveness of transformations in each domain for improving adversarial transferability.

Adversarial examples are generated on Vgg-16 and tested on other models. The attack success rates on each dataset are reported in Fig. 6. First, we observe that individual transformations in either frequency or spatial domains can enhance adversarial transferability, resulting in a higher attack success rate. Besides, compared to the frequency domain, more significant enhancements can be achieved by performing transformations in the spatial domain. This indicates the priority of enhancing target-related diversity over decoupling object and background when generating adversarial examples. Finally, incorporating transformations in the spatial and frequency domains into SFCoT achieves the best attack performance.

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TABLE V
ABLATION EXPERIMENTS ABOUT THE INFLUENCE OF PERFORMING TRANSFORMATION ON DIFFERENT FREQUENCY COMPONENTS

Components	Vgg-16	Vgg-19	Res-34	Res-50
ALL	44.8%	46.5%	58.5%	63.1%
HH	26.4%	29.2%	38.8%	40.3%
HL&LH	27.5%	29.5%	40.6%	42.0%
LL	**45.3%**	**47.4%**	**58.8%**	**64.7%**



(a) (b)

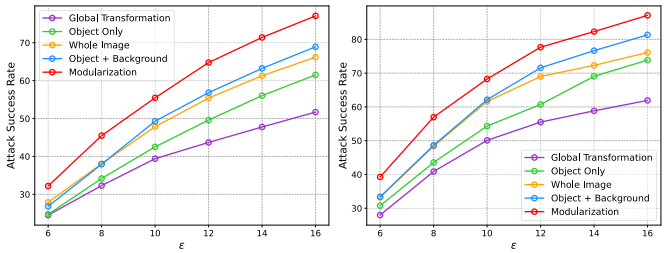
Fig. 6. Ablation experiments for transformations in each domain. (a) FGSCR-42. (b) MTARSI.

2) Selection of Frequency Components: We selectively perform transformations on low-frequency components to narrow the gap between different models. To confirm our motivation in frequency component selection, we investigate the influence of performing frequency transformations on different components decomposed by DWT using the MTARSI dataset.

The attack performance of adversarial examples generated by Res-34 on different models is reported in Table V. The ALL represents performing transformations on components of all four subbands, i.e., global transformation. According to the result, frequency transformations on solely high or medium-high frequency components do not enhance attack performance, but cause severe performance degradation compared to the global transformation instead. In contrast, selectively performing transformation on low-frequency components can consistently improve the adversarial transferability across different target models. The above analysis convincingly confirms our motivation: performing transformations on low-frequency components can effectively improve attack efficacy for remote sensing object recognition.

3) Modularization of Spatial Blocks: We modularly apply various spatial transformations to different image blocks, simulating target-related input patterns in changes of observation and environment. In this experiment, we compare the transferability of adversarial examples generated using different spatial transformation strategies. Specifically, we vary the constraint on perturbation from 6 to 16 when employing Vgg-16 as the source model to generate adversarial examples on the MTARSI dataset.

The success attack rates against Res-34 and Dense-121 are shown in Fig. 7. The Global Transformation corresponds to unspecified global spatial transformations of DIM. The Object Only refers to performing specified transformations solely on the divided target-related block. The Whole Image refers to performing identical transformations to the whole image rather than to blocks. The Object + Background differs from our adopted MST in that all environment-related blocks



(a) (b)

Fig. 7. Ablation experiments for different spatial transformation strategies. (a) Res-34. (b) Dense-121.

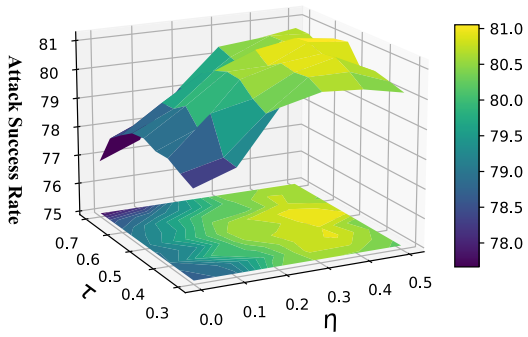


Fig. 8. Attack success rate of adversarial examples generated by MST with different combinations of τ and η.

are transformed uniformly. First, compared to globally applying common spatial transformations, purposefully enhancing target-related diversity from the perspective of observation effectively improves adversarial transferability. Then, applying content-specified transformations to different regions of the image can better mitigate overfitting to model-specific information than applying identical transformations to the whole image. Finally, employing more diverse transformations in modules can comprehensively cover potential input patterns, thus achieving the best attack performance in the black-box setting.

D. Parameter Analysis

In this section, we analyze the parameter settings involved in the transformation process of SFCoT, including the activation threshold τ, the dimension range η, the scaling amplitude δ, and size N of the transformed image set. The experiments are carried out on the MTARSI dataset using Res-34 as the source model.

First, we evaluate the attack performance with different combinations of τ and η, whose values vary from 0.3 to 0.7 and 0 to 0.5, respectively. The average attack success rates against different target models (with only spatial transformations) are shown in Fig. 8. We can see that the performance improves with the increase in η over a certain range, but it saturates or tends to degrade when η becomes too large. In comparison, the attack performance remains stable with changes in τ. Then, regarding transformations in the frequency domain, we vary δ from 0 to 0.4. According to the results shown in Fig. 9(a), the proposed method is sensitive to changes in δ. Adversarial transferability improves with amplitude scaling in the frequency domain, but excessive scaling can cause

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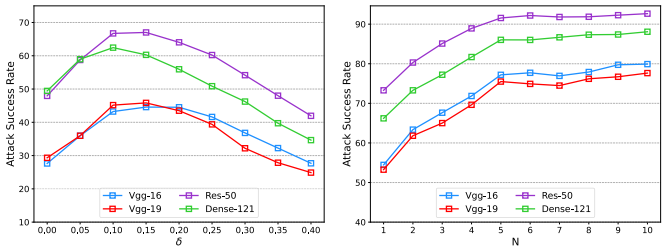


Fig. 9. Attack success rate with different settings of (a) δ and (b) N.

the opposite effect. Finally, we investigate the impact of N on SFCoT during gradient calculation, as shown in Fig. 9(b). The attack success rate continuously increases with N, but the incremental gain gradually decreases. With consideration of computational efficiency, taking a medium size for N is more reasonable.

V. Conclusion

In this article, we have proposed the SFCoT for adversarial attacks against remote sensing object recognition. The proposed SFCoT incorporates transformations in both spatial and frequency domains, effectively enhancing transferability across models. Specifically, transformations are purposefully applied to different compositions rather than globally to the entire image. In the frequency domain, we selectively perform transformations on low-frequency components to narrow intermodel gaps. Subsequently, spatial transformations are modularly applied to enhance target-related diversity. Finally, the average gradient is calculated over the set of transformed images, which comprehensively covers the potential diversity of input patterns, thereby stably generating adversarial examples in iterations. Extensive evaluations have been conducted to prove the effectiveness of SFCoT, which outperforms state-of-the-art methods across models. Apart from the superiority in performance, the established benchmarks for black-box attacks against remote sensing object recognition also provide deep insights into model weakness. In future research, we aim to delve into adversarial learning in the field of remote sensing, encompassing both adversarial attacks and defenses. In particular, we will investigate the impact of adversarial attacks on model robustness across different data modalities and implement robust defense mechanisms through the effective fusion of multimodal data.

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